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Inventor(s)	Khedekar; Amol V. et al.

Housing of an HVAC system

Abstract

A frame segment for a frame of a heating, ventilation, and/or air conditioning (HVAC) system includes a protrusion formed on an interior side of the frame segment and configured to engage with a first panel of the HVAC system, a first recess formed at an exterior side, opposite the interior side, of the frame segment, and a second recess formed at the exterior side of the frame segment. The first recess is configured to receive a second panel of the HVAC system, and the second recess is configured to receive a third panel of the HVAC system.

Inventors: Khedekar; Amol V. (Pune, IN), Amick; Robert A. (York, PA), Nagime; Shivchandra V. (Latur, IN), Urey; Henry L. (York, PA), Honnamane; Mahantesh (Pune, IN)

Applicant: Johnson Controls Tyco IP Holdings LLP (Milwaukee, WI)

Family ID: 1000008819574

Assignee: Tyco Fire & Security GmbH (Neuhausen am Rheinfall, CH)

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Primary Examiner: Jules; Frantz F

Assistant Examiner: Tadesse; Martha

Attorney, Agent or Firm: Fletcher Yoder, PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority from and the benefit of India Provisional Application No. 202011042331, entitled “MULLION STRUCTURE FOR AN AIR HANDLING UNIT,” filed Sep. 29, 2020, which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

(2) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure and are described below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be noted that these statements are to be read in this light, and not as admissions of prior art.

(3) Heating, ventilation, and/or air conditioning (HVAC) systems are utilized in residential, commercial, and industrial environments to control environmental properties, such as temperature and humidity, for occupants of the respective environments. An HVAC system may control the environmental properties through control of a supply air flow delivered to the environment. For example, the HVAC system may place the supply air flow in a heat exchange relationship with a refrigerant of a vapor compression circuit to condition the supply air flow. In certain embodiments, the HVAC system may include a housing or an enclosure, which may include segments, panels, connectors, and so forth that may be assembled to define an interior volume of the HVAC system.

Unfortunately, traditional housings may be difficult to manufacture in a desirable manner, such as to include a desirable arrangement of compartments or sections. Thus, manufacture of an HVAC system may be difficult, expensive, and/or time consuming, and/or a functionality of the HVAC system may be limited.

SUMMARY

(4) A summary of certain embodiments disclosed herein is set forth below. It should be noted that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(5) In one embodiment, a frame segment for a frame of a heating, ventilation, and/or air conditioning (HVAC) system includes a protrusion formed on an interior side of the frame segment and configured to engage with a first panel of the HVAC system, a first recess formed at an exterior side, opposite the interior side, of the frame segment, and a second recess formed at the exterior side of the frame segment. The first recess is configured to receive a second panel of the HVAC system, and the second recess is configured to receive a third panel of the HVAC system.

(6) In one embodiment, a frame structure of a heating, ventilation, and/or air conditioning (HVAC) system includes a frame segment having a protrusion formed on a first side of the frame segment and having a first recess and a second recess formed on a second side, opposite the first side, of the frame segment. The frame structure also includes a frame connector configured to couple to the frame segment.

(7) In one embodiment, a housing for a heating, ventilation, and/or air conditioning (HVAC) system includes a frame structure having a frame segment. The frame segment includes a protrusion formed on a first side of the frame segment, a first recess formed on a second side, opposite the first side, of the frame segment, and a second recess formed on the second side of the frame segment. The housing also includes a plurality of housing panels. A first housing panel of the plurality of housing panels is configured to couple to the protrusion of the frame segment, a second housing panel of the plurality of housing panels is configured to be disposed within the first recess of the frame segment, and a third housing panel of the plurality of housing panels is configured to be disposed within the second recess of the frame segment

Description

DRAWINGS

(1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:

(2) FIG. 1 is a perspective view of a building including a heating, ventilating, or air conditioning (HVAC) system, in accordance with an aspect of the present disclosure;

(3) FIG. 2 is a block diagram of an airside system including an air handling unit (AHU) which can be used in the HVAC system of FIG. 1, in accordance with an aspect of the present disclosure;

(4) FIG. 3 is a perspective view of an embodiment of an HVAC system having a housing, in accordance with an aspect of the present disclosure;

(5) FIG. 4 is a perspective view of an embodiment of a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(6) FIG. 5 is a detailed exploded view of an embodiment of a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(7) FIG. 6 is a detailed perspective view of an embodiment of a frame segment configured to be incorporated in a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(8) FIG. 7 is an axial view of an embodiment of a frame segment configured to be incorporated in a

frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(9) FIG. 8 is a detailed perspective view of an embodiment of a frame segment configured to be incorporated in a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(10) FIG. 9 is an axial view of an embodiment of a frame segment configured to be incorporated in a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(11) FIG. 10 is a detailed perspective view of an embodiment of a frame segment configured to be incorporated in a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(12) FIG. 11 is an axial view of an embodiment of a frame segment configured to be incorporated in a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(13) FIG. 12 is a detailed perspective view of an embodiment of a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(14) FIG. 13 is a detailed perspective view of an embodiment of a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(15) FIG. 14 is a perspective view of an embodiment of a connector configured to be incorporated in a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure;

(16) FIG. 15 is a detailed perspective view of an embodiment of a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure; and

(17) FIG. 16 is a perspective view of an embodiment of a frame structure of a housing for an HVAC system, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION

(18) One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be noted that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(19) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be noted that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(20) The present disclosure is directed to a heating, ventilation, and/or air conditioning (HVAC) system. The HVAC system may be configured to condition an air flow and to deliver the air flow to a space serviced by the HVAC system to condition the space. As an example, the HVAC system may include a vapor compression circuit configured to circulate a refrigerant, cool the refrigerant via a condenser, place the cooled refrigerant in a heat exchange relationship with the air flow to cool the air flow (e.g., via an evaporator), and deliver the air flow to the space to cool the space. As another example, the HVAC system may be configured to heat the air flow, such as via a furnace, and deliver the air flow to the space to heat the space.

(21) The HVAC system may include a housing or an enclosure configured to contain various components and devices (e.g., the condenser, the evaporator) and shield the components from an exterior environment, such as an ambient environment. For example, the housing may include a frame structure and panels that cooperatively define an interior volume in which the components may be disposed. However, the frame structure for housings of traditional or conventional design may not be easily arranged, configured, and/or modified to define the interior volume as desired, such as to form a particular number of sections or compartments and/or sections or compartments of a particular size. Thus, the arrangement of the frame structure may have various limitations, which may limit certain functionalities of the HVAC system. Indeed, it may be difficult to manufacture a housing with multiple sections using components of conventional frame structures.

(22) Thus, it is presently recognized that improvements associated with manufacture of the housing are needed. Accordingly, embodiments of the present disclosure are directed to connectors (e.g., frame connectors) and segments (e.g., frame segments) configured to couple to one another to form a frame structure that may define multiple sections of the housing in an assembled configuration. For example, multiple peripheral frame assemblies defining respective housing sections may be configured to couple to an intermediate frame assembly of the frame structure. Each of the peripheral frame assemblies may define a respective housing section. Thus, the frame structure may include multiple housing sections that are integral with one another in an assembled configuration. In some embodiments, the intermediate frame assembly may include intermediate frame connectors and intermediate frame segments configured to couple to the intermediate frame connectors, and the peripheral frame assembly may include peripheral frame connectors and peripheral frame segments configured to couple to both the peripheral frame connectors and the intermediate frame connectors. Each of the intermediate frame segments may have a profile to enable the intermediate frame segment to couple to at least three panels (e.g., housing panels) of the HVAC system. Additionally, each of the intermediate frame connectors may have a profile to enable the intermediate frame connector to couple to multiple peripheral frame segments and multiple intermediate frame segments. The arrangement of the frame structure to define multiple housing sections integral with one another may improve manufacture and/or operation of HVAC systems. By way of example, one of the housing sections may include an air handling unit (AHU) and another of the housing sections may include a vestibule. The air handling unit may enclose different components that enable operation of the HVAC system to condition an air flow, and the integral coupling between the AHU and the vestibule may enable certain components of the AHU to be accessible to a user (e.g., a technician, an operator) via the vestibule. For example, the vestibule may be an enclosed housing section that defines a protected interior space configured to accommodate an operator or technician and enable access to components of the AHU. In this manner, the frame structure may improve access to various components of the HVAC system, such as for servicing the HVAC system, thereby improving performance of the HVAC system. However, it should be noted that the frame structure may enable manufacture of an HVAC system having housing sections for enabling any suitable functionality.

(23) Referring now to FIG. 1, a perspective view of a building **10** is shown. The building **10** is served by a heating, ventilating, or air conditioning (HVAC) system **100**. The HVAC system **100** can include a plurality of HVAC devices (e.g., heaters, chillers, air handling units, pumps, fans, thermal energy storage) configured to provide heating, cooling, air conditioning, ventilation, and/or other services for the building **10**. For example, the HVAC system **100** is shown to include a waterside system **120** and an airside system **130**. The waterside system **120** may provide a heated or chilled fluid to an air handling unit of the airside system **130**. The airside system **130** may use the heated or chilled fluid to heat or cool an airflow provided to the building **10**.

(24) The HVAC system **100** is shown to include a chiller **102**, a boiler **104**, and a rooftop AHU **106**. The waterside system **120** may use the boiler **104** and the chiller **102** to heat or cool a working fluid (e.g., water, glycol) and may circulate the working fluid to the AHU **106**. In various embodiments,

the HVAC devices of the waterside system **120** can be located in or around the building **10** (as shown in FIG. **1**) or at an offsite location such as a central plant (e.g., a chiller plant, a steam plant, a heat plant) that serves one or more buildings including the building **10**. The working fluid can be heated in the boiler **104** or cooled in the chiller **102**, depending on whether heating or cooling is required in the building **10**. The boiler **104** may add heat to the circulated fluid, for example, by burning a combustible material (e.g., natural gas) or using an electric heating element. The chiller **102** may place the circulated fluid in a heat exchange relationship with another fluid (e.g., a refrigerant) in a heat exchanger (e.g., an evaporator) to absorb heat from the circulated fluid. The working fluid from the chiller **102** and/or the boiler **104** can be transported to the AHU **106** via piping **108**.

(25) The AHU **106** may place the working fluid in a heat exchange relationship with an airflow passing through the AHU **106** (e.g., via one or more stages of cooling coils and/or heating coils). The airflow can be, for example, outside air, return air from within the building **10**, or a combination of both. The AHU **106** may transfer heat between the airflow and the working fluid to provide heating or cooling for the airflow. For example, the AHU **106** can include one or more fans or blowers configured to pass the airflow over or through a heat exchanger containing the working fluid. The working fluid may then return to the chiller **102** or the boiler **104** via piping **110**.

(26) The airside system **130** may deliver the airflow supplied by the AHU **106** (i.e., the supply airflow) to the building **10** via air supply ducts **112** and may provide return air from the building **10** to the AHU **106** via air return ducts **114**. In some embodiments, the airside system **130** includes multiple variable air volume (VAV) units **116**. For example, the airside system **130** is shown to include a separate VAV unit **116** on each floor or zone of the building **10**. The VAV units **116** can include dampers or other flow control elements that can be operated to control an amount of the supply airflow provided to individual zones of the building **10**. In other embodiments, the airside system **130** delivers the supply airflow into one or more zones of the building **10** (e.g., via the supply ducts **112**) without using intermediate VAV units **116** or other flow control elements. The AHU **106** can include various sensors (e.g., temperature sensors, pressure sensors) configured to measure attributes of the supply airflow. The AHU **106** may receive input from sensors located within the AHU **106** and/or within the building zone and may adjust the flow rate, temperature, or other attributes of the supply airflow through the AHU **106** to achieve setpoint conditions for the building zone.

(27) Referring now to FIG. **2**, a block diagram of an airside system **200** is shown, according to some embodiments. In various embodiments, the airside system **200** may supplement or replace the airside system **130** in the HVAC system **100** or can be implemented separate from the HVAC system **100**. When implemented in the HVAC system **100**, the airside system **200** can include a subset of the HVAC devices in the HVAC system **100** (e.g., AHU **106**, VAV units **116**, ducts **112-114**, fans, dampers) and can be located in or around the building **10**. The airside system **200** may operate to heat or cool an airflow provided to the building **10** using a heated or chilled fluid provided by the waterside system **120**.

(28) In FIG. **2**, the airside system **200** is shown to include an AHU **202**. Economizer-type AHUs vary the amount of outside air and return air used by the air handling unit for heating or cooling. For example, the AHU **202** may receive return air **204** from a building zone **206** via a return air duct **208** and may deliver supply air **210** to the building zone **206** via a supply air duct **212**. In some embodiments, the AHU **202** is a rooftop unit located on the roof of the building **10** (e.g., the AHU **106** as shown in FIG. **1**) or otherwise positioned to receive both the return air **204** and outside air **214**. The AHU **202** can be configured to operate an exhaust air damper **216**, a mixing damper **218**, and an outside air damper **220** to control an amount of the outside air **214** and the return air **204** that combine to form the supply air **210**. Any return air **204** that does not pass through the mixing damper **218** can be exhausted from the AHU **202** through the exhaust damper **216** as exhaust air **222**.

(29) Each of the dampers **216-220** can be operated by an actuator. For example, the exhaust air damper **216** can be operated by an actuator **224**, the mixing damper **218** can be operated by an actuator **226**, and the outside air damper **220** can be operated by an actuator **228**. The actuators **224-228** may communicate with an AHU controller **230** via a communications link **232**. The actuators **224-228** may receive control signals from the AHU controller **230** and may provide feedback signals to the AHU controller **230**. Feedback signals can include, for example, an indication of a current actuator or damper position, an amount of torque or force exerted by the actuator, diagnostic information (e.g., results of diagnostic tests performed by the actuators **224-228**), status information, commissioning information, configuration settings, calibration data, and/or other types of information or data that can be collected, stored, or used by the actuators **224-228**. The AHU controller **230** can be an economizer controller configured to use one or more control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms) to control the actuators **224-228**.

(30) Still referring to FIG. 2, the AHU **202** is shown to include a cooling coil **234**, a heating coil **236**, and a fan **238** positioned within the supply air duct **212**. The fan **238** can be configured to force the supply air **210** through the cooling coil **234** and/or the heating coil **236** and provide the supply air **210** to the building zone **206**. The AHU controller **230** may communicate with the fan **238** via a communications link **240** to control a flow rate of the supply air **210**. In some embodiments, the AHU controller **230** controls an amount of heating or cooling applied to the supply air **210** by modulating a speed of the fan **238**.

(31) The cooling coil **234** may receive a chilled fluid from the waterside system **120** (via piping **242**) and may return the chilled fluid to the waterside system **120** via piping **244**. A valve **246** can be positioned along the piping **242** or the piping **244** to control a flow rate of the chilled fluid through the cooling coil **234**. In some embodiments, the cooling coil **234** includes multiple stages of cooling coils that can be independently activated and deactivated (e.g., by the AHU controller **230**, by a supervisory controller **266**) to modulate an amount of cooling applied to the supply air **210**.

(32) The heating coil **236** may receive a heated fluid from the waterside system **120** via piping **248** and may return the heated fluid to the waterside system **120** via piping **250**. A valve **252** can be positioned along the piping **248** or the piping **250** to control a flow rate of the heated fluid through the heating coil **236**. In some embodiments, the heating coil **236** includes multiple stages of heating coils that can be independently activated and deactivated (e.g., by the AHU controller **230**, by the supervisory controller **266**) to modulate an amount of heating applied to the supply air **210**.

(33) Each of the valves **246**, **252** can be controlled by an actuator. For example, the valve **246** can be controlled by an actuator **254**, and the valve **252** can be controlled by an actuator **256**. The actuators **254**, **256** may communicate with the AHU controller **230** via communications links **258**, **260**. The actuators **254** and **256** may receive control signals from the AHU controller **230** and may provide feedback signals to the AHU controller **230**. In some embodiments, the AHU controller **230** receives a measurement of the supply air temperature from a temperature sensor **262** positioned in the supply air duct **212** (e.g., downstream of the cooling coil **234** and/or the heating coil **236**). The AHU controller **230** may also receive a measurement of the temperature of the building zone **206** from a temperature sensor **264** located in the building zone **206**.

(34) In some embodiments, the AHU controller **230** operates the valves **246** and **252** via the actuators **254**, **256** to modulate an amount of heating or cooling provided to the supply air **210** (e.g., to achieve a setpoint temperature for the supply air **210** or to maintain the temperature of the supply air **210** within a setpoint temperature range). The positions of the valves **246**, **252** affect the amount of heating or cooling provided to the supply air **210** by the cooling coil **234** or the heating coil **236** and may correlate with the amount of energy consumed to achieve a desired supply air

temperature. The AHU controller **230** may control the temperature of the supply air **210** and/or the building zone **206** by activating or deactivating the coils **234**, **236**, adjusting a speed of the fan **238**, or a combination of both. While the illustrated embodiment is configured to enable heating and cooling of the supply air **210** via fluids received from the waterside system **120**, it should be appreciated that additional or alternative embodiments of the AHU **202** may include a vapor compression circuit (e.g., a refrigerant circuit), a furnace, and/or other components configured to heat and/or cool the supply air **210**.

(35) Still referring to FIG. 2, the airside system **200** is shown to include the supervisory controller **266** and a client device **268**. The supervisory controller **266** can include one or more computer systems (e.g., servers, supervisory controllers, subsystem controllers) that serve as system level controllers, application or data servers, head nodes, or master controllers for the airside system **200**, the waterside system **120**, the HVAC system **100**, and/or other controllable systems that serve the building **10**. The supervisory controller **266** may communicate with multiple downstream building systems or subsystems (e.g., the HVAC system **100**, a security system, a lighting system, the waterside system **120**) via a communications link **270** according to like or disparate protocols (e.g., LON, BACnet). In various embodiments, the AHU controller **230** and the supervisory controller **266** can be separate (as shown in FIG. 2) or integrated. In an integrated implementation, the AHU controller **230** can be a software module configured for execution by processing circuitry of the supervisory controller **266**.

(36) In some embodiments, the AHU controller **230** receives information from the supervisory controller **266** (e.g., commands, setpoints, operating boundaries) and provides information to the supervisory controller **266** (e.g., temperature measurements, valve or actuator positions, operating statuses, diagnostics). For example, the AHU controller **230** may provide the supervisory controller **266** with temperature measurements from the temperature sensors **262**, **264**, equipment on/off states, equipment operating capacities, and/or any other information that can be used by the supervisory controller **266** to monitor or control a variable state or condition within the building zone **206**.

(37) The client device **268** can include one or more human-machine interfaces or client interfaces (e.g., graphical user interfaces, reporting interfaces, text-based computer interfaces, client-facing web services, web servers that provide pages to web clients) for controlling, viewing, or otherwise interacting with the HVAC system **100**, its subsystems, and/or devices. The client device **268** can be a computer workstation, a client terminal, a remote or local interface, or any other type of user interface device. The client device **268** can be a stationary terminal or a mobile device. For example, the client device **268** can be a desktop computer, a computer server with a user interface, a laptop computer, a tablet, a smartphone, a PDA, or any other type of mobile or non-mobile device. The client device **268** may communicate with the supervisory controller **266** and/or the AHU controller **230** via communications link **272**.

(38) The present disclosure is directed to a housing of an HVAC system, such as a housing of the HVAC system **100**. The housing may include a frame structure having frame connectors and frame segments coupled to one another to define multiple housing sections of the HVAC system, such as housing sections that are integrated with, directly coupled to, or extending from one another. The profile of the frame connectors and frame segments may increase flexibility, ease, and/or efficiency of manufacture of the housing, such as to define a suitable number and/or size of the housing sections. Thus, the frame structure described herein may improve the manufacture and/or functionality of the HVAC system.

(39) With this in mind, FIG. 3 is a perspective view of an HVAC system **300** that includes a housing or enclosure **302** having a frame structure **304** (e.g., frame, a housing frame, an enclosure frame). In the illustrated embodiment, the frame structure **304** includes a first housing section **306**, which may be an AHU (e.g., the AHU **106**), and a second housing section **308**, which may be a vestibule. Each of the housing sections **306**, **308** may define a respective interior volume within the

housing **302**, and the housing sections **306**, **308** may be integral with (e.g., directly coupled to) one another. Furthermore, the housing **302** may include a plurality of housing panels **310** coupled to the frame structure **304** and configured to shroud or enclose the interior volumes of the housing sections **306**, **308** from an external (e.g., ambient) environment. For instance, the housing panels **310** may shield the interior volumes from external elements, such as a temperature, precipitation, debris, and so forth.

(40) By way of example, the first housing section **306** may include, contain, and/or enclose components, such as a compressor, an evaporator, a condenser, a burner, a fan, and so forth, disposed in the interior volume and configured to operate to condition an air flow directed through the first housing section **306**. In some embodiments, the first housing section **306** may define the supply air duct **212** and may include the cooling coil **234**, the heating coil **236**, and/or the fan **238** discussed above with reference to FIG. 2. Indeed, the first housing section **306** may be fluidly coupled to ductwork (e.g., the air supply ducts **112**, the air return ducts **114**) to enable the air flow to be directed from the housing **302** into a space and/or from the space into the housing **302**. For example, the first housing section **306** may include an opening **311** that may be configured to fluidly couple to the ductwork. The second housing section **308** may enable access to one or more of the components disposed within the first housing section **306**. For instance, an interior panel, an interior wall, an interior partition, an interior divider, and/or an interior door may be disposed within the housing **302** (e.g., coupled to the frame structure **304**) between the first housing section **306** and the second housing section **308**.

(41) The second housing section **308** may enable a user to adjust (e.g., open) the interior panel to access the interior volume of the first housing section **306**. For example, the second housing section **308** may include an external door **312** (e.g., access panel) that is adjustable to enable access to the interior volume of the second housing section **308**. Upon opening the external door **312**, the user may access the interior volume of the second housing section **308** and thereby access the interior panel within the housing **302** to access the interior volume of the first housing section **306**. For instance, the external door **312** may enable access to the interior volume of the first housing section **306** while the housing panels **310** remain attached at the first housing section **306**. In this way, the user may access the components disposed within the first housing section **306** while such components are shielded by the housing panels **310** and are in operation. The second housing section **308** may also shield the user from certain elements of the external environment (e.g., during inclement weather) while the user accesses any of the interior volumes within the housing **302**. As an example, a size of the interior volume defined by the second housing section **308** may at least partially accommodate the user. As such, the illustrated housing **302** may help the user perform various tasks associated with the HVAC system **300**, such as maintenance and service of the HVAC system **300**. In some embodiments, certain components of the HVAC system **300** may also be disposed within the second housing section **308**. By way of example, the second housing section **308** may be configured to enclose and shield electronic components (e.g., wiring), piping or coil connections (e.g., connections to components disposed within the first housing section **306**), and the like. Thus, the user may also open the external door **312** to access such components disposed in the second housing section **308**.

(42) It should be noted that in additional or alternative embodiments, the first housing section **306** and the second housing section **308** may include any suitable section of the HVAC system **300**. For example, the housing sections **306**, **308** may be used for any suitable purpose and/or may contain any suitable components of the HVAC system **300**. Further still, in one or more embodiments, the interior panel may not be disposed between the housing sections **306**, **308** or otherwise may not fully separate the first housing section **306** and the second housing section **308** from one another. Thus, the respective interior volumes of the first housing section **306** and the second housing section **308** may remain at least partially continuous with one another.

(43) FIG. 4 is a perspective view of an embodiment of the frame structure **304** of the housing **302**.

To facilitate discussion, the frame structure **304** and its components will be described with reference to a longitudinal axis or direction **330**, a vertical axis or direction **332**, and a lateral axis or direction **334** with respect to an orientation of the HVAC system **300**. The frame structure **304** may include an intermediate frame assembly **336** (e.g., intermediate assembly) configured to couple adjacent peripheral frame assemblies **338** with one another to define the housing sections **306**, **308** and the housing **302**. As an example, the intermediate frame assembly **336** may include a first intermediate frame segment or mullion **340** (e.g., a top frame segment, a horizontal frame segment), a second intermediate frame segment or mullion **342** (e.g., a front frame segment, a downstream frame segment, a last-in-airstream frame segment, a vertical frame segment), a third intermediate frame segment or mullion **344** (e.g., a bottom frame segment, a horizontal frame segment), and a fourth intermediate frame segment or mullion **346** (e.g., a rear frame segment, an upstream frame segment, a first-in-airstream segment, a vertical frame segment). Each of the intermediate frame segments **340**, **342**, **344**, **346** may be configured to couple to one another via intermediate frame connectors **348** (e.g., mullion connectors). That is, the intermediate frame segments **340**, **342**, **344**, **346** may be fastened, secured, or otherwise attached (e.g., directly) to one or more intermediate frame connectors **348**. For example, the first intermediate frame segment **340** may be coupled to the second intermediate frame segment **342** via a first intermediate frame connector **348**, the second intermediate frame segment **342** may be coupled to the third intermediate frame segment **344** via a second intermediate frame connector **348**, the third intermediate frame segment **344** may be coupled to the fourth intermediate frame segment **346** via a third intermediate frame connector **348**, and the fourth intermediate frame segment **346** may be coupled to the first intermediate frame segment **340** via a fourth intermediate frame connector **348**.

(44) Each of the peripheral frame assemblies **338** may include a plurality of peripheral frame segments or raceways **350** coupled to one another via peripheral frame connectors **352**. The intermediate frame assembly **336** and the peripheral frame assemblies **338** may cooperatively define the housing sections **306**, **308**. In the illustrated embodiment, each of the peripheral frame connectors **352** is coupled to three peripheral frame segments **350** extending crosswise to one another (e.g., along the axes **330**, **332**, **334**). Additionally, each of the intermediate frame connectors **348** is configured to couple to a pair of peripheral frame segments **350** (e.g., extending along the lateral axis **334**), such as respective peripheral frame segments **350** partially defining each of the housing sections **306**, **308**. In this manner, the intermediate frame assembly **336** may also partially define both the first housing section **306** and the second housing section **308**.

(45) As further discussed herein, each of the frame segments **340**, **342**, **344**, **346**, **350** may have a profile (e.g., geometry, configuration) configured to enable coupling to one or more of the housing panels **310**. Additionally, in an assembled configuration of the intermediate frame assembly **336**, the intermediate frame segments **340**, **342**, **344**, **346** may be configured to cooperatively receive (e.g., accommodate, align with) the interior panel disposed between the housing sections **306**, **308** to separate the housing sections **306**, **308** from one another. In some embodiments, each of the intermediate frame segments **340**, **342**, **344**, **346** may have a different profile (e.g., geometry, configuration) than that of the peripheral frame segments **350**. Furthermore, a subset of the intermediate frame segments (e.g., the first intermediate frame segment **340** and the third intermediate frame segment **344**) may have a different profile than that of another subset of the intermediate frame segments (e.g., the second intermediate frame segment **342** and the fourth intermediate frame segment **346**). Such profiles may have features and/or dimensions to facilitate coupling the frame segments **340**, **342**, **344**, **346**, **350**, the frame connectors **348**, **352**, and the housing panels **310** with one another.

(46) The frame structure **304** may enable greater flexibility and configurability to form any suitable number of housing sections for the HVAC system **300**. For example, in certain embodiments, the frame structure **304** may include multiple intermediate frame assemblies **336** to define more than two housing sections **306**, **308**. An embodiment having two intermediate frame assemblies **336** may

include peripheral frame segments **350** that couple to and extend between respective intermediate frame connectors **348** of the intermediate frame assemblies **336**. Furthermore, differently sized frame segments **340, 342, 344, 346, 350** may be selected to define a size of the interior volumes of the housing sections of the housing **302**. For example, shorter frame segments may be selected to reduce a dimension of any of the interior volumes, longer frame segments may be selected to increase a dimension of any of the interior volumes, and so forth. Indeed, the first housing section **306** may be larger than the second housing section **308**, the first housing section **306** may be smaller than the second housing section **308**, or the first housing section **306** and the second housing section **308** may be approximately the same size. While the frame segments **340, 342, 344, 346, 350** may have different lengths, frame segments **340, 342, 344, 346, 350** of a common type (e.g., intermediate frame segments **340, 344**, intermediate frame segments **342, 346**, or peripheral frame segment **350**) may still have a common profile, shape or configuration. Thus, a common design or configuration of each type of frame segment **340, 342, 344, 346, 350** may be manufactured and then cut according to a desired size (e.g., length). In this way, the frame segments **340, 342, 344, 346, 350** and the frame connectors **348, 352** may increase flexibility and efficiency associated with manufacture and/or assembly of the frame structure **304**. That is, multiple different embodiments of the housing **302** may be manufactured via the frame segments **340, 342, 344, 346, 350** and frame connectors **348, 352**. Further still, although the illustrated housing **302** has a rectangular geometry, the HVAC system **300** may have any suitably shaped housing **302**, such as any suitably shaped frame structure **304** and/or housing sections **306, 308** formed from the frame segments **340, 342, 344, 346, 350** and frame connectors **348, 352**.

(47) FIG. 5 is a detailed, exploded view of an embodiment of the frame structure **304** of the housing **302**. The illustrated embodiment includes one of the intermediate frame connectors **348** configured to couple to the first intermediate frame segment **340** the second intermediate frame segment **342**, and two of the peripheral frame segments **350**. By way of example, the intermediate frame connector **348** may include a bracket **380** that has a first peripheral extension **382** extending from a first side **384** (e.g., a first lateral side) of the intermediate frame connector **348** and a second peripheral extension **386** extending from a second side **388** (e.g., a second lateral side), opposite the first side **384**, of the intermediate frame connector **348**. The peripheral extensions **382, 386** may be configured to couple to respective peripheral frame segments **350**. For example, each of the peripheral frame segments **350** may include (e.g., define) a first opening, space, or passage **390** extending therethrough, and the peripheral extensions **382, 386** may be inserted into the respective first openings **390** of the peripheral frame segments **350**. In some embodiments, the peripheral extensions **382, 386** may have a geometry, shape, and/or dimension corresponding to at least a portion of a geometry, shape, or dimension of the peripheral frame segments **350** and/or the first openings **390**. Additionally, first holes **392** may be formed through the peripheral extensions **382, 386**. In an assembled configuration of the frame structure **304**, in which the peripheral extensions **382, 386** are inserted into the first openings **390** of the peripheral frame segments **350**, the first holes **392** of the peripheral extensions **382, 386** may align with second holes **394** formed in the peripheral frame segments **350**. Mechanical fasteners (not shown), such as bolts, rivets, and the like, may be inserted through the aligned holes **392, 394** to bias the peripheral extensions **382, 386** and the respective peripheral frame segments **350** against one another. Thus, the mechanical fasteners may secure the first peripheral extension **382** to one of the peripheral frame segments **350** and the second peripheral extension **386** to another of the peripheral frame segments **350**, thereby coupling the peripheral frame segments **350** to the intermediate frame connector **348**.

(48) The bracket **380** of the intermediate frame connector **348** may also include a first intermediate extension **396** extending from a third side **398** of the intermediate frame connector **348** and a second intermediate extension **400** extending from a fourth side **402**, opposite the third side **398**, of the intermediate frame connector **348**. The first intermediate extension **396** may be configured to couple to the first intermediate frame segment **340**, and the second intermediate extension **400** may

be configured to couple to the second intermediate frame segment **342**. As an example, the first intermediate frame segment **340** may form a second opening, space, or channel **404** extending therethrough, and the first intermediate extension **396** may be inserted into the second opening **404**. Additionally, third holes **406** may be formed through the first intermediate extension **396**, and fourth holes **408** may be formed through the first intermediate frame segment **340**. The third holes **406** and the fourth holes **408** may align with one another in the assembled configuration, and mechanical fasteners may be inserted through the aligned holes **406**, **408** to bias the first intermediate frame segment **340** and the first intermediate extension **396** against one another, thereby securing the first intermediate frame segment **340** to the intermediate frame connector **348**. Furthermore, the second intermediate frame segment **342** may form a third opening, space, or channel **410** extending therethrough, and the second intermediate extension **400** may be inserted into the third opening **410**. Fifth holes **412** may be formed through the second intermediate extension **400**, and sixth holes **414** may be formed through the second intermediate frame segment **342**. The fifth holes **412** and the sixth holes **414** may align with one another in the assembled configuration, and mechanical fasteners may be inserted through the aligned holes **406**, **408** to bias the second intermediate frame segment **342** and the second intermediate extension **400** against one another, thereby securing the second intermediate frame segment **342** to the intermediate frame connector **348**.

(49) In certain embodiments, the intermediate frame connector **348** may also include seventh holes **416** formed on the peripheral extensions **382**, **386** (e.g., on a respective surface that is oriented crosswise with respect to the corresponding surfaces on which the first holes **392** are formed). The seventh holes **416** may also enable the peripheral frame segments **350** to be coupled to the intermediate frame connector **348**, such as for a different relative orientation between the intermediate frame connector **348** and the peripheral frame segments **350**. By way of example, the intermediate frame connector **348** may be arranged in a different manner at another location on the intermediate frame assembly **336**. For instance, in another position and orientation of the intermediate frame connector **348**, the first holes **392** may be occluded (e.g., by a base of the HVAC system **300** or housing **302**) and the seventh holes **416** may be exposed. For this reason, at such a position and orientation of the intermediate frame connector **348**, it may not be desirable to arrange the peripheral frame segments **350** in a manner that enables the second holes **394** of the peripheral frame segments **350** to align with the first holes **392** of the peripheral extensions **382**, **386** in the assembled configuration. Instead, the peripheral frame segments **350** may be oriented to enable the second holes **394** of the peripheral frame segments **350** to align with the seventh holes **416** of the peripheral extensions **382**, **386** to facilitate insertion of the mechanical fasteners through the aligned holes **394**, **416** to couple the peripheral frame segments **350** to the intermediate frame connector **348**. Forming the first holes **392** and the seventh holes **416** through each of the peripheral extensions **382**, **386** may enable flexible and selectable orientation and positioning of the intermediate frame connector **348** during assembly the frame structure **304**. As an example, a single embodiment of the intermediate frame connector **348** may be manufactured and incorporated in any suitable manner (e.g., any position, any orientation) within the intermediate frame assembly **336**, and the peripheral frame segments **350** may be coupled to any of the intermediate frame connectors **348**.

(50) In some embodiments, the intermediate frame connector **348** may include a shroud **418** (e.g., shroud portion, cover portion) configured to couple to the bracket **380** (e.g., to exterior surfaces of the bracket **380**). As an example, the shroud **418** may be coupled to the bracket **380** via a mechanical fastener, an adhesive, a punch, a weld, another suitable feature, or any combination thereof. The shroud **418** may be configured to provide a sealing interface for the intermediate frame connector **348**. For instance, in the assembled configuration, each of the frame segments **340**, **342**, **350** may be configured to engage the shroud **418**. Indeed, respective edges of the frame segments **340**, **342**, **350** may be configured to abut corresponding edges of the shroud **418**. Thus, the shroud

418 may occlude gaps formed between the frame segments **340, 342, 350** and/or the bracket **380** in the assembled configuration, thereby blocking air from flowing between an exterior and an interior of the housing **302**. In this way, the shroud **418** may enable the HVAC system **300** to operate more efficiently.

(51) Additionally, port holes **420** may be formed through the bracket **380** and/or the shroud **418**. The port holes **420** may enable material to be injected or inserted through the intermediate frame connector **348**, and the material may flow from the intermediate frame connector **348** and into the openings **390, 404, 410** to fill the frame segments **340, 342, 350** in the assembled configuration. For example, the material may include an insulating material that limits heat transfer between the interior of the housing **302** and the external environment. Thus, the material may block an impact of a temperature of the external environment on the components disposed within the housing **302** and/or air flow directed through the housing **302** to enable the HVAC system **300** to operate more efficiently.

(52) In the illustrated embodiment, the extensions **382, 386, 396, 400** of the intermediate frame connector **348** are oriented approximately perpendicularly relative to one another. In additional or alternative embodiments, the extensions **382, 386, 396, 400** may be oriented at different angles than depicted, such as at an oblique angle and/or at an approximately parallel angle. In further embodiments, the intermediate frame connector **348** may include a different number of extensions **382, 386, 396, 400** than depicted. Further still, although the illustrated intermediate frame connector **348** is configured to couple to the frame segments **340, 342, 350**, the intermediate frame connector **348** may be configured to couple to different frame segments, such as to at least one of the intermediate frame segments **344, 346**.

(53) In some embodiments, the first intermediate frame segment **340** and the third intermediate frame segment **344** may have a common profile, shape, configuration, and/or geometry. That is, the intermediate frame segments **340, 344** may be of a single embodiment or design. FIG. 6 is a detailed perspective view of an embodiment of the intermediate frame segment **340, 344** (e.g., horizontal intermediate frame segment). To facilitate discussion, the intermediate frame segment **340, 344** will be described with reference to a longitudinal axis or direction **440**, a vertical axis or direction **442**, and a lateral axis or direction **444** with respect to an orientation of the intermediate frame segment **340, 344**. The intermediate frame segment **340, 344** may include an interior side **446**, which may face an interior of the housing **302** in the assembled configuration of the frame structure **304**. The intermediate frame segment **340, 344** may also include an exterior side **448**, which may face an exterior of the housing **302** and/or an environment surrounding the housing **302** in the assembled configuration. The intermediate frame segment **340, 344** may further include a first lateral side **450** and a second lateral side **452**.

(54) A protrusion **454** may be formed at the interior side **446** of the intermediate frame segment **340, 344**. The protrusion **454** may include a first protrusion portion **456** extending from a first interior surface **458**, a second protrusion portion **460** extending from a second interior surface **462**, and a third protrusion portion **464** extending between the first and second protrusion portions **456, 460**. Thus, the first protrusion portion **456** and the second protrusion portion **460** may offset an interior protrusion surface **466** of the third protrusion portion **464** from the first interior surface **458** and the second interior surface **462**. The protrusion **454** may be configured to couple to a panel, such as the interior panel, in the assembled configuration of the frame structure **304**. By way of example, an end of the panel may be configured to couple (e.g., mount) to a lateral protrusion surface **468** of the first protrusion portion **456** of one of the intermediate frame segments **340, 344** (e.g., via mechanical fasteners). An opposite end of the panel may be configured to couple (e.g., mount) to a corresponding lateral protrusion surface **468** of the first protrusion portion **456** of another of the intermediate frame segments **340, 344** (e.g., disposed at an opposite side of the frame structure **304**). In other words, the respective lateral protrusion surfaces **468** of the intermediate frame segments **340, 344** may couple to opposite ends of the panel. Additionally, an interior edge

470 extending along the protrusion **454** (e.g., along the protrusion portions **456**, **460**, **464**) and along the first interior surface **458** may engage a corresponding edge of one of the intermediate frame segments **342**, **346** in the assembled configuration. Thus, air may be blocked from flowing between the assembled intermediate frame segments **340**, **342**, **344**, **346** (e.g., to block air from flowing between an interior and an exterior of the housing **302**).

(55) The exterior side **448** of the intermediate frame segment **340**, **344** may include a first exterior portion **472** extending from a first exterior surface **474**, a second exterior portion **476** extending from a second exterior surface **478**, and a third exterior portion **480** extending between the first and second exterior portions **472**, **476**. Thus, the first exterior portion **472** and the second exterior portion **476** may offset a third exterior surface **482** from the first exterior surface **474** and the second exterior surface **478**. In this manner, the orientation of the exterior portions **472**, **476**, **480** relative to the exterior surfaces **474**, **478** may form recesses **484** at the exterior side **448** and extending along opposite lateral sides **450**, **452** (e.g., along the longitudinal axis **440**). In the assembled configuration, each of the recesses **484** may be configured to receive at least a portion of one of the housing panels **310**. For example, the first exterior surface **474** and the second exterior surface **478** may be configured to couple to separate housing panels **310** (e.g., via mechanical fasteners). In some embodiments, each of the exterior surfaces **474**, **478** may generally extend along a plane formed by the longitudinal axis **440** and the lateral axis **444** and may therefore be oriented crosswise to the vertical axis **442** and/or the lateral protrusion surface **468**. As such, the housing panels **310** disposed at least partially within the recesses **484** may be oriented crosswise to the panel (e.g., interior panel) coupled to the protrusion **454**. Additionally, the fourth holes **408** may be formed through the third exterior portion **480** to enable the third exterior portion **480** to be coupled to the intermediate frame connector **348** (e.g., the first intermediate extension **396**) in the assembled configuration. In the assembled configuration, an exterior edge **486** extending along the exterior portions **472**, **476**, **480** may engage the shroud **418** of the intermediate frame connector **348** and/or corresponding edges of the peripheral frame segments **350**, thereby providing a sealing interface therebetween to block air from flowing between the intermediate frame segments **340**, **344**, the peripheral frame segments **350**, and the intermediate frame connector **348**.

(56) The first lateral side **450** may include a first lateral portion **488** extending between the first exterior surface **474** and the second interior surface **462**. In the illustrated embodiment, a first flange **490** extends beyond (e.g., outwardly from, relative to the second opening **404**) the first lateral portion **488** along the lateral axis **444**. The first flange **490** may be configured to engage a corresponding flange of a peripheral frame segment **350** (e.g., a peripheral frame segment **350** engaging the first lateral side **450** of the intermediate frame segment **340**, **344**) in the assembled configuration, further blocking air from flowing between the intermediate frame segment **340**, **344** and the peripheral frame segment **350** at the first lateral side **450**. In certain embodiments, a first flange surface **491** of the first flange **490** may be configured to engage with another component of the HVAC system **300** and/or housing **302** in the assembled configuration, such as one of the housing panels **310**, a base panel, a cover, or any other suitable component. The second lateral side **452** may include a second lateral portion **492** extending between the first interior surface **458** and the second exterior surface **478**. A second flange **494** may extend beyond (e.g., outwardly from, relative to the second opening **404**) the second lateral portion **492** along the lateral axis **444**. The second flange **494** may be configured to engage a corresponding flange of an additional peripheral frame segment **350** (e.g., a peripheral frame segment **350** engaging the second lateral side **452** of the intermediate frame segment **340**, **344**) in the assembled configuration, further blocking air from flowing between the intermediate frame segment **340**, **344** and the additional peripheral frame segment **350** at the second lateral side **452**. A second flange surface **495** of the second flange **494** may be configured to engage another component of the HVAC system **300** and/or housing **302** in the assembled configuration, as similarly described above.

(57) In the illustrated embodiment, first lateral edges **496** (e.g., first interior edges) extending (e.g.,

from the interior edge **470**, from the exterior edge **486**) along the first exterior surface **474**, the second interior surface **462**, and the first flange **490** may be angled obliquely relative to the lateral axis **444**. In this way, the first lateral edges **496** may be configured to receive and engage with corresponding angled edges of the peripheral frame segment **350** engaging the first lateral side **450** of the intermediate frame segment **340**, **344**. Similarly, second lateral edges **498** (e.g., second interior edges) extending along the second exterior surface **478** and the second flange **494** may be angled obliquely relative to the lateral axis **444**. Thus, the second lateral edges **498** may be configured to receive and engage corresponding angled edges of the peripheral frame segment **350** engaging the second lateral side **452** of the intermediate frame segment **340**, **344**.

(58) The intermediate frame segment **340**, **344** may also include a projection **500** extending from the second lateral portion **492** and the second flange **494** at the interior side **446** and in a direction that is at least partially along the longitudinal axis **440**. That is, the projection **500** may be oriented obliquely relative to the second lateral edges **498**. In the assembled configuration, the projection **500** may extend at least partially into the first opening **390** of the peripheral frame segment **350** engaging the second lateral side **452** of the intermediate frame segment **340**, **344** and may engage a surface of the peripheral frame segment **350** located within the first opening **390**. In this way, the projection **500** may further block air from flowing between the intermediate frame segment **340**, **344** and the peripheral frame segment **350**. The projection **500** may also increase securement between the intermediate frame segment **340**, **344** and the peripheral frame segment **350**. For instance, the projection **500** may block certain relative movement between the intermediate frame segment **340**, **344** and the peripheral frame segment **350** (e.g., along the vertical axis **442**). Although the illustrated projection **500** has a rectangular shape, an additional or alternative projection **500** may include any suitable geometry, such as a triangular shape, a curved shape, a wavy shape, and so forth.

(59) FIG. 7 is an axial view of an embodiment of the intermediate frame segment **340**, **344**. Each of the first exterior portion **472** and the second exterior portion **476** may include a respective first inward facing surface **520** (e.g., a lateral inward facing surface that is exposed to the second opening **404**). Additionally, the third exterior portion **480** may include a second inward facing surface **522** (e.g., a surface that is exposed to the second opening **404** and extends between the first inward facing surfaces **520**). In the assembled configuration, each of the first inward facing surfaces **520** and the second inward facing surface **522** may be configured to engage a portion of the intermediate frame connector **348**. For example, the inward facing surfaces **520**, **522** may form a channel **524** in the second opening **404** at the exterior side **448**. The channel **524** may capture the first intermediate extension **396** to block relative movement between the intermediate frame segment **340**, **344** and the intermediate frame connector **348**.

(60) In the illustrated embodiment, a length of the first protrusion portion **456** is greater than a length of the second protrusion portion **460**. That is, with respect to the vertical axis **442**, an offset (e.g., dimension) between the third interior protrusion surface **466** and the first interior surface **458** may be greater than an offset (e.g., dimension) between the third protrusion surface **466** and the second interior surface **462**. Thus, the first interior surface **458** may be offset from the second interior surface **462** relative to the vertical axis **442**. For example, the offset between the interior surfaces **458**, **462** may enable the projection **500** extending from the second lateral portion **492** to be positioned within the first opening **390** of the peripheral frame segment **350** engaging the second lateral side **452** of the intermediate frame segment **340**, **344**.

(61) Furthermore, in certain embodiments, the second intermediate frame segment **342** and the fourth intermediate frame segment **346** may have generally the same profile. Indeed, the intermediate frame segments **342**, **346** may be of a single embodiment that is different than that of the first and second intermediate frame segments **340**, **344**. FIG. 8 is a detailed perspective view of an embodiment of the intermediate frame segment **342**, **346** (e.g., vertical frame segment). To facilitate discussion, the intermediate frame segment **342**, **346** will be described with reference to a

longitudinal axis or direction **550**, a vertical axis or direction **552**, and a lateral axis or direction **554** with respect to an orientation of the intermediate frame segment **342**, **346**. The intermediate frame segment **342**, **346** may include an interior side **556**, which may face an interior of the housing **302** in the assembled configuration, and an exterior side **558**, which may face an exterior of the housing **302** in the assembled configuration. The intermediate frame segment **342**, **346** may also include a first lateral side **560** and a second lateral side **562**.

(62) A protrusion **564** may be formed at the interior side **556**. The protrusion **564** may include a first protrusion portion **566** extending from a first interior surface **568**, a second protrusion portion **570** extending from a second interior surface **572**, and a third protrusion portion **574** extending between the first and second protrusion portions **566**, **570**. The first protrusion portion **566** and the second protrusion portion **570** may offset an interior protrusion surface **576** of the third protrusion portion **574** from the first interior surface **568** and the second interior surface **572**. The protrusion **564** may be configured to couple to a panel, such as the interior panel that is also coupled to the protrusion **454** of the intermediate frame segment **340**, **344**, in the assembled configuration. Indeed, the second protrusion portion **570** may include a lateral protrusion surface **578** configured to couple (e.g., mount) to an end of the panel. Additionally, an opposite end of the panel may be configured to couple to a corresponding lateral protrusion surface **578** of the first protrusion portion **566** of another of the intermediate frame segments **342**, **346**. Thus, respective lateral protrusion surfaces **578** of the intermediate frame segments **342**, **346** may couple to opposite ends of the panel. The interior side **556** may also include an extended portion **580**, which may extend beyond a remainder of intermediate frame segment **342**, **346** along the longitudinal axis **550**. That is, with respect to the longitudinal axis **550**, a first interior edge **582** extending along the protrusion **564** (e.g., along the protrusion portions **566**, **570**, **574**) may be offset from a second interior edge **584** extending along the extended portion **580**. In the assembled configuration, each of the interior edges **582**, **584** may engage one of the intermediate frame segments **340**, **344** (e.g., horizontal intermediate frame segments). For example, the first interior edge **582** of the intermediate frame segment **342**, **346** may engage the interior protrusion surface **466** of the intermediate frame segment **340**, **344**, and the second interior edge **584** of the intermediate frame segment **342**, **346** may engage the first interior surface **458** of the intermediate frame segment **340**, **344**, thereby blocking air from flowing between the assembled intermediate frame segments **340**, **342**, **344**, **346**.

(63) The exterior side **558** of the intermediate frame segment **342**, **346** may include a first exterior portion **586** extending from a first exterior surface **588**, a second exterior portion **590** extending from a second exterior surface **592**, and a third exterior portion **594** extending between the first and second exterior portions **586**, **590**. Thus, the first exterior portion **586** and the second exterior portion **590** may offset a third exterior surface **596** from the first exterior surface **588** and the second exterior surface **592**. The orientation of the exterior portions **586**, **590**, **594** relative to the exterior surfaces **588**, **592** may form recesses **598** at the exterior side **558** and extending along the opposing lateral sides **560**, **562** (e.g., along the longitudinal axis **550**). In the assembled configuration, each of the recesses **598** may be configured to receive (e.g., accommodate) one of the housing panels **310**, which may couple to the exterior surfaces **588**, **592**, for example. For instance, the exterior surfaces **588**, **592** may be oriented crosswise to the lateral protrusion surface **578**. Thus, the housing panels **310** at least partially disposed within the recesses **598** may be oriented crosswise to the panel (e.g., interior panel) coupled to the protrusion **564**. Furthermore, the sixth holes **414** may be formed through the third exterior portion **594** to enable the third exterior portion **594** to be coupled to the intermediate frame connector **348** (e.g., the second intermediate extension **400**) in the assembled configuration. Further still, in the assembled configuration, an exterior edge **600** extending along the exterior portions **586**, **590**, **594** may engage the shroud **418** of the intermediate frame connector **348** and/or corresponding edges of the peripheral frame segments **350**, thereby blocking air from flowing between the intermediate frame segments **342**, **346**, the peripheral frame segments **350**, and the intermediate frame connector **348**.

(64) The first lateral side **560** may include a first lateral portion **602** extending between the first interior surface **568** and the first exterior surface **588** to form a first lateral channel **603** within the third opening **410** of the intermediate frame segment **342, 346**. A first lateral edge **604** extending along the first lateral channel **603** may engage the peripheral frame segment **350** positioned at the first lateral side **560** of the intermediate frame segment **342, 346**. The second lateral side **562** may include a second lateral portion **606** extending between the second interior surface **572** and the second exterior surface **592** to form a second lateral channel **605** within the third opening **410** of the intermediate frame segment **342, 346**. A second lateral edge **608** extending along the second lateral channel **605** may engage the peripheral frame segment **350** positioned at the second lateral side **562** of the intermediate frame segment **342, 346**. Thus, the lateral portions **602, 606** may further block air from flowing between the intermediate frame segment **342, 346** and the peripheral frame segments **350**. The lateral portions **602, 606** may also block relative movement between the intermediate frame segment **342, 346** and the peripheral frame segments **350**.

(65) FIG. **9** is an axial view of an embodiment of the intermediate frame segment **342, 346**. Each of the first exterior portion **586** and the second exterior portion **590** may include a respective first inward facing surface **630** (e.g., a lateral inward facing surface that is exposed to the third opening **410**). Additionally, the third exterior portion **594** may include a second inward facing surface **632** (e.g., a surface that is exposed to the third opening **410** and extends between the first inward facing surfaces **630**). In the assembled configuration, each of the first inward facing surfaces **630** and the second inward facing surface **632** may be configured to engage a portion of the intermediate frame connector **348**. By way of example, the inward facing surfaces **630, 632** may form a channel **634** in the third opening **410** at the exterior side **558**. The channel **634** may capture the second intermediate extension **400** to block relative movement between the intermediate frame segment **342, 346** and the intermediate frame connector **348**.

(66) In some embodiments, each of the peripheral frame segments **350** may have a common profile and be of the same embodiment. FIG. **10** is a detailed perspective view of an embodiment of the peripheral frame segment **350**. To facilitate discussion, the peripheral frame segment **350** will be described with reference to a longitudinal axis or direction **650**, a vertical axis or direction **652**, and a lateral axis or direction **654** with respect to an orientation of the peripheral frame segment **350**. The peripheral frame segment **350** may include a first side **656** (e.g., a first exterior side), a second side **658** (e.g., a second exterior side), a third side **660** (e.g., a first interior side), and a fourth side **662** (e.g., a second interior side).

(67) The first side **656** may include a first surface **664**, a second surface **666**, and a first portion **668** that offsets the first surface **664** and the second surface **666** from one another along the vertical axis **652** to form a first recess **669**. In the assembled configuration, the first recess **669** may be configured to at least partially receive one of the housing panels **310**, which may be coupled to the second surface **666**, and the first portion **668** may be configured to engage one of the exterior portions **472, 476** of the intermediate frame segments **340, 344**. The second side **658** may include a third surface **670**, a fourth surface **672**, a second portion **674** that offsets the third surface **670** and the fourth surface **672** from one another along the lateral axis **654** to form a second recess **675**, and a third portion **676** extending from the second portion **674**. In the assembled configuration, the second recess **675** may receive one of the housing panels **310**, which may be coupled to the fourth surface **672**, for example. A first edge **680** extending along the first portion **668**, the first surface **664**, the third surface **670**, the second portion **674**, and the third portion **676** may engage the shroud **418** of the intermediate frame connector **348** in the assembled configuration, thereby blocking air from flowing between the intermediate frame connector **348** and the peripheral frame segment **350**. Moreover, the second holes **394** may be formed through the third surface **670** to enable the peripheral frame segment **350** to be coupled to the intermediate frame connector **348**.

(68) The third side **660** may include a fifth surface **684**, a sixth surface **686**, and a fourth portion **688** that offsets the fifth surface **684** and the sixth surface **686** from one another along the vertical

axis **652**. In the assembled configuration, the fifth surface **684** may engage one of the intermediate frame segments **342, 346** (e.g., the first lateral edge **604**, the second lateral edge **608**), thereby blocking air from flowing between the peripheral frame segment **350** and the intermediate frame segments **342, 346**. Additionally, in the assembled configuration, the fourth portion **688** may be configured to engage one of the extended portion **580** of the intermediate frame segments **342, 346** or the second protrusion portion **460** of the intermediate frame segments **340, 344**. A portion of a second edge **689** extending along the fifth surface **684** and the fourth portion **688** may also engage the shroud **418** of the intermediate frame connector **348** in the assembled configuration, further blocking air from flowing between the intermediate frame connector **348** and the peripheral frame segment **350**. The fourth side **662** may include a fifth portion **690** extending between the second surface **666** and the sixth surface **686**. A flange **692** may extend beyond (e.g., outwardly from) the fifth portion **690** along the lateral axis **654**. The flange **692** may be configured to engage a corresponding flange of another frame segment (e.g., the first flange **490** or the second flange **494** of the intermediate frame segment **340, 344**).

(69) Further, in the illustrated embodiment, third edges **694** (e.g., third interior edges) extending (e.g., from the first edge **680**, from the second edge **689**) along the second surface **666**, the sixth surface **686**, and the flange **692** may be angled obliquely relative to the lateral axis **654**. Thus, the third edges **694** may be configured to receive corresponding angled edges of the intermediate frame segments **340, 344** (e.g., the first lateral edges **496**, the second lateral edges **498**). Moreover, the fifth portion **690** of the peripheral frame segment **350** may engage the first lateral portion **488** or the second lateral portion **492** of the intermediate frame segments **340, 344**. Such interfaces between the peripheral frame segment **350** and the intermediate frame segments **340, 344** may block air from flowing between the peripheral frame segment **350** and the intermediate frame segments **340, 344**.

(70) FIG. **11** is an axial view of an embodiment of the peripheral frame segment **350**. The peripheral frame segment **350** may include a first inward facing surface **720** (e.g., opposite the first surface **664**) and a second inward facing surface **722** (e.g., opposite the third surface **670**). Each of the inward facing surfaces **720, 722** may be exposed to the first opening **390** and may be configured to engage a portion of the intermediate frame connector **348**. For example, the inward facing surfaces **720, 722** may cooperatively capture one of the peripheral extensions **382, 386** to block relative movement between the peripheral frame segment **350** and the intermediate frame connector **348**.

(71) In the illustrated embodiment, a length of the third portion **676** is greater than a length of the fourth portion **688**. That is, with respect to the vertical axis **652**, an offset between the fifth surface **684** and a seventh surface **723** extending along the second portion **674** may be greater than an offset between the fifth surface **684** and the sixth surface **686**. For example, the offset between the fifth surface **684** and the sixth surface **686** may enable the projection **500** of the intermediate frame segment **340, 344** to extend into the first opening **390**, such as to engage the third inward facing surface **724** (e.g., opposite the sixth surface **686**), in the assembled configuration.

(72) Although each of the illustrated frame segments **340, 342, 344, 346, 350** includes rectangular features (e.g., portions that are oriented generally perpendicularly to one another), any of the frame segments **340, 342, 344, 346, 350** may have features of any other suitable geometry in additional or alternative embodiments. Moreover, in some embodiments, any of the intermediate frame segments **340, 342, 344, 346, 350** may include components that are separately manufactured (e.g., via metal bending, via extrusion) and are coupled to one another, such as via mechanical fasteners, adhesives, welding, brazing, and the like. In additional or alternative embodiments, any of the frame segments **340, 342, 344, 346, 350** may be formed from a single integral component (e.g., bending and cutting a single piece of material). Indeed, each of the frame segments **340, 342, 344, 346, 350** may be formed to contain insulation material injected into their respective openings **390, 404, 410**, such as to provide sealed interfaces and/or boundaries that block the insulation material from being

undesirably removed from an interior of the frame segments **340, 342, 344, 346, 350** (e.g., to the exterior of the housing **302**). In other words, the openings **390, 404, 410** formed by the intermediate frame segments **340, 342, 344, 346, 350** may be sealed to contain the insulation material therein.

(73) FIG. 12 is a detailed perspective view of an embodiment of the frame structure **304**, illustrating two peripheral frame segments **350**, one of the intermediate frame segments **340, 344** (e.g., horizontal intermediate frame segments), and one of the intermediate frame segments **342, 346** (e.g., vertical intermediate frame segments) coupled to one another. In the illustrated embodiment, the intermediate frame connector **348** is not shown to better illustrate engagement between the frame segments **340, 342, 344, 346, 350**. The illustrated intermediate frame segment **340, 344** is engaged with a first peripheral frame segment **350A** at the first lateral side **450** of the intermediate frame segment **340, 344**. For instance, the first exterior portion **472** of the intermediate frame segment **340, 344** may be configured to engage with the first portion **668** (not shown) of the first peripheral frame segment **350A**, the first lateral edges **496** of the intermediate frame segment **340, 344** may be configured to engage with the third edges **694** of the first peripheral frame segment **350A**, the first lateral portion **488** of the intermediate frame segment **340, 344** may be configured to engage with the fifth portion **690** (not shown) of the first peripheral frame segment **350A**, the first flange **490** of the intermediate frame segment **340, 344** may be configured to engage with the flange **692** (not shown) of the first peripheral frame segment **350A**, and the second protrusion portion **460** (not shown) of the intermediate frame segment **340, 344** may be configured to engage with the fourth portion **688** (not shown) of the first peripheral frame segment **350A**.

(74) Additionally, the intermediate frame segment **340, 344** is engaged with a second peripheral frame segment **350B** at the second lateral side **452** of the intermediate frame segment **340, 344**. By way of example, the second exterior portion **476** of the intermediate frame segment **340, 344** may be configured to engage with the first portion **668** of the second peripheral frame segment **350B**, the second lateral edges **498** of the intermediate frame segment **340, 344** may be configured to engage with the third edges **694** of the second peripheral frame segment **350B**, the second lateral portion **492** of the intermediate frame segment **340, 344** may be configured to engage with the fifth portion **690** of the second peripheral frame segment **350B**, the second flange **494** (not shown) of the intermediate frame segment **340, 344** may be configured to engage with the flange **692** (not shown) of the second peripheral frame segment **350B**, and the projection **500** of the intermediate frame segment **340, 344** may be configured to insert into the first opening **390** and engage with the third inward facing surface **724** of the second peripheral frame segment **350B**.

(75) The intermediate frame segment **340, 344** is also engaged with the intermediate frame segment **342, 346**. For instance, the protrusion **454** (e.g., the interior protrusion surface **466** of the intermediate frame segment **340, 344**) may be configured to engage with the protrusion **564** (e.g., the first interior edge **582**) of the intermediate frame segment **342, 346**, and each of the first protrusion portion **456**, the interior edge **470**, and the first interior surface **458** of the intermediate frame segment **340, 344** may be configured to engage with the extended portion **580** of the intermediate frame segment **342, 346** (e.g., the second interior edge **584**, the second interior surface **572**). The intermediate frame segment **342, 346** is further engaged to the peripheral frame segments **350**. As an example, the first lateral edge **604** of the intermediate frame segment **342, 346** may be configured to engage with the fifth surface **684** of the first peripheral frame segment **350A**, and the second lateral edge **608** of the intermediate frame segment **342, 346** may be configured to engage with the fifth surface **684** of the second peripheral frame segment **350B**.

(76) Housing panels **310** may be positioned along various surfaces of the frame structure **304** in the illustrated embodiment. For instance, a respective housing panel **310** may be configured to be coupled to the interface formed by the first exterior surface **474** of the intermediate frame segment **340, 344** and the second surface **666** of the first peripheral frame segment **350A**, the interface

formed by the first flange **490** of the intermediate frame segment **340, 344** and the flange **692** of the first peripheral frame segment **350A**, the interface formed by the second exterior surface **478** of the intermediate frame segment **340, 344** and the second surface **666** of the second peripheral frame segment **350B**, the interface formed by the second flange **494** of the intermediate frame segment **340, 344** and the flange **692** of the second peripheral frame segment **350B**, the interface formed by the first exterior surface **588** of the intermediate frame segment **342, 346** and the fourth surface **672** of the first peripheral frame segment **350A**, and/or the interface formed by the second exterior surface **592** of the intermediate frame segment **342, 346** and the fourth surface **672** of the second peripheral frame segment **350B**.

(77) FIG. **13** is a detailed perspective view of an embodiment of the frame structure **304** at an interior of the housing **302**. The illustrated frame structure **304** includes the two peripheral frame segments **350**, one of the intermediate frame segments **340, 344** (e.g., horizontal intermediate frame segments), and one of the intermediate frame segments **342, 346** (e.g., vertical intermediate frame segments) coupled to one another. FIG. **13** more clearly illustrates the engagement between the first flange **490** of the intermediate frame segment **340, 344** and the flange **692** of the first peripheral frame segment **350A** and the engagement between the second flange **494** of the intermediate frame segment **340, 344** and the flange **692** of the second peripheral frame segment **350B**. The illustrated embodiment also includes an interface formed by the lateral protrusion surface **468** of the intermediate frame segment **340, 344** and the lateral protrusion surface **578** of the intermediate frame segment **342, 346**. One of the housing panels **310** (shown in phantom lines), such as the interior panel, may be coupled to this interface in the assembled configuration.

(78) FIG. **14** is a perspective view of an embodiment of the intermediate frame connector **348**. To facilitate discussion, the intermediate frame connector **348** will be described with reference to a longitudinal axis **750**, a vertical axis **752**, and a lateral axis **754** with respect to an orientation of the intermediate frame connector **348**. The illustrated intermediate frame connector **348** includes the bracket **380** having a first portion **756** and a second portion **758** that extend crosswise to one another. The first portion **756** may include the first intermediate extension **396** and may also form a part of the peripheral extensions **382, 386**. The second portion **758** may include the second intermediate extension **400** and may also form a part of the peripheral extensions **382, 386**. The first holes **392** may be formed in the peripheral extensions **382, 386** of the second intermediate extension **400**, and the seventh holes **416** may be formed in the peripheral extensions **382, 386** of the first portion **756**.

(79) The first portion **756** may include a first base **760** extending along the lateral axis **754**. A first end **762** of the first base **760** may form part of the first peripheral extension **382**, and a second end **764**, opposite the first end **762**, of the first base **760** may form part of the second peripheral extension **386**. A first support **766** may extend (e.g., along the vertical axis **752**) from the first base **760** (e.g., a medial section of the first base **760**) to form the first intermediate extension **396**. Thus, the first base **760** and the first support **766** may collectively have a T-shaped geometry.

Additionally, first flanges **768** may extend from the first support **766** (e.g., along the longitudinal axis **750**). Each of the first support **766** and the first flanges **768** may be captured by one of the intermediate frame segments **340, 344** (e.g., within the second opening **404**) to block movement between the intermediate frame connector **348** and the intermediate frame segment **340, 344**. One of the port holes **420** may also be formed through a medial section of the first base **760**.

(80) The second portion **758** may include a second base **770** extending along the lateral axis **754** (e.g., along the first base **760**). The second base **770** may be oriented crosswise to the first base **760**. A third end **772** of the second base **770** may form another part of the first peripheral extension **382**, and a fourth end **774**, opposite the third end **772**, may form another part of the second peripheral extension **386**. A second support **776** may extend (e.g., along the longitudinal axis **750**) from the second base **770** (e.g., a medial section of the second base **770**) to form the second intermediate extension **400**. As such, the second base **770** and the second support **776** may

collectively have a T-shaped geometry. Second flanges **778** may extend from the second support **776** (e.g., along the vertical axis **752**). Each of the second support **776** and the second flanges **778** may be captured by one of the intermediate frame segments **342**, **346** (e.g., within the third opening **410**), thereby blocking movement between the intermediate frame connector **348** and the intermediate frame segment **342**, **346**. Another of the port holes **420** may be formed through a medial section of the second base **770**.

(81) In certain embodiments, the portions **756**, **758** may be integrally formed with one another. For instance, the portions **756**, **758** may be formed from a single piece of material by cutting, metal bending, extrusion, and/or from multiple material pieces integrally secured to one another by welding or another joining technique. In additional or alternative embodiments, the portions **756**, **758** may be separate components that are coupled to one another, such as via a mechanical fastener or adhesive.

(82) The shroud **418** may be coupled to the bracket **380** along exterior surfaces **780** (e.g., surfaces that are exterior relative to an internal angle **782** formed between the bases **760**, **770**) of the first base **760**, the first support **766**, the second base **770**, and the second support **776**. For instance, a mechanical fastener, an adhesive, a weld, a punch, an interference fit, and/or any other suitable feature may be used to couple the shroud **418** to the bracket **380**.

(83) FIG. **15** is a perspective view of an embodiment of the frame structure **304** at an interior of the housing **302**. The illustrated frame structure **304** includes one of the intermediate frame segments **340**, **344**, one of the intermediate frame segments **342**, **346**, and two peripheral frame segments **350**. A portion of the intermediate frame segment **340**, **344** and of the intermediate frame segment **342**, **346** is not shown for visualization purposes. The intermediate frame connector **348** is coupled to each of the intermediate frame segment **340**, **344**, the intermediate frame segment **342**, **346**, and the peripheral frame segments **350**. For example, the first peripheral extension **382** is configured to extend into the first opening **390** of the first peripheral segment **350A**, the second peripheral extension **386** (not shown) is configured to extend into the first opening **390** of the second peripheral segment **350B**, the first intermediate extension **396** is configured to extend into the second opening **404** of the intermediate frame segment **340**, **344**, and the second intermediate extension **400** is configured to extend into the third opening **410** of the intermediate frame segment **342**, **346**.

(84) In the illustrated embodiment, one of the first flanges **768** of the first intermediate extension **396** is configured to abut the first inward facing surface **520** of the first exterior portion **472** of the intermediate frame segment **340**, **344**. The other of the first flanges **768** of the first intermediate extension **396** may abut the first inward facing surface **520** of the second exterior portion **476** of the intermediate frame segment **340**, **344**. Thus, the exterior portions **472**, **476** may capture the first intermediate extension **396**, thereby blocking movement between the intermediate frame segment **340**, **344** and the intermediate frame connector **348**. Furthermore, one of the second flanges **778** of the second intermediate extension **400** is configured to abut the first inward facing surface **630** of the first exterior portion **586** of the intermediate frame segment **342**, **346**, and the other of the second flanges **778** is configured to abut the first inward facing surface **630** of the second exterior portion **590** of the intermediate frame segment **342**, **346**. As such, the exterior portions **586**, **590** may capture the second intermediate extension **400**, thereby blocking movement between the intermediate frame segment **342**, **346** and the intermediate frame connector **348**.

(85) In the assembled configuration, the frame segments **340**, **342**, **344**, **346**, **350** may be fluidly coupled to one another via the intermediate frame connector **348**. Indeed, each of the openings **390**, **404**, **410** may be configured to receive material (e.g., insulation material) inserted through the port hole **420**. Thus, the material inserted through the port hole **420** may be used to fill the openings **390**, **404**, **410** extending through the respective frame segments **340**, **342**, **344**, **346**, **350**.

(86) FIG. **16** is a perspective view of an embodiment of a frame structure **800** for the housing **302** of the HVAC system **300**. The illustrated frame structure **800** includes four housing sections **802**,

804, 806, 808. In some embodiments, the frame structure **800** may include multiple frame structures **304** described above coupled to one another. By way of example, a subset of the peripheral frame segments **350**, the second intermediate frame segments **342**, and the intermediate frame connector **348** of one of the frame structures **304** may be coupled to a subset of the peripheral frame segments **350**, the second intermediate frame segments **342**, and the intermediate frame connector **348** of another of the frame structures **304**. For instance, a tie or bind (e.g., a cable tie), a mechanical fastener, an adhesive, a weld, another suitable feature, or any combination thereof may couple the components of the individual frame structures **304** to one another. Indeed, any suitable number of frame structures **304** may be coupled to one another to form the frame structure **800** having a desirable number of housing sections (e.g., more than four sections). In additional or alternative embodiments, different components may be used to create a frame structure having a desirable number of housing sections without coupling individual intermediate frame connectors **348** and/or peripheral frame segments **350** to one another. For example, a single intermediate frame connector may be configured to couple to multiple first intermediate frame segments **340**, multiple second intermediate frame segments **342**, multiple third intermediate frame segments **344**, and/or multiple fourth intermediate frame segments **346**, thereby enabling multiple intermediate frame assemblies **336** to be incorporated in the frame structure.

(87) The present disclosure may provide one or more technical effects useful in an HVAC system. For example, the HVAC system may include a housing with a frame structure. In some embodiments, the frame structure may include an intermediate frame assembly that may be configured to couple multiple peripheral frame assemblies to one another. The intermediate frame assembly may include an intermediate frame connector configured to couple to multiple intermediate frame segments of the intermediate frame assembly and multiple peripheral frame segments of the respective peripheral frame assemblies. Each of the peripheral frame assemblies may define a respective housing section of the HVAC system with the intermediate frame assembly. Thus, the intermediate frame assembly may enable the frame structure to form multiple housing sections that are integral with or directly coupled to one another. Each housing section may be used for a different purpose associated with the HVAC system. As an example, one of the housing sections may contain components for operating a vapor-compression system of the HVAC system, and another housing section may be used to facilitate maintenance of the vapor-compression system. In this manner, the intermediate frame assembly may increase a functionality and/or improve manufacture of the HVAC system. The technical effects and technical problems in the specification are examples and are not limiting. It should be noted that the embodiments described in the specification may have other technical effects and can solve other technical problems.

(88) While only certain features and embodiments of the disclosure have been illustrated and described, many modifications and changes may occur to those skilled in the art, such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, including temperatures and pressures, mounting arrangements, use of materials, colors, orientations, and so forth without materially departing from the novel teachings and advantages of the subject matter recited in the claims. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure. Furthermore, in an effort to provide a concise description of the exemplary embodiments, all features of an actual implementation may not have been described, such as those unrelated to the presently contemplated best mode of carrying out the disclosure, or those unrelated to enabling the claimed disclosure. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation specific decisions may be made. Such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those

of ordinary skill having the benefit of this disclosure, without undue experimentation.

(89) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

Claims

1. A housing for a heating, ventilation, and/or air conditioning (HVAC) system, comprising: a frame structure comprising a frame segment, wherein the frame segment comprises a protrusion formed on a first side of the frame segment, a first recess formed on a second side, opposite the first side, of the frame segment, and a second recess formed on the second side of the frame segment; and a plurality of housing panels, wherein a first housing panel of the plurality of housing panels is configured to directly couple to the protrusion of the frame segment, a second housing panel of the plurality of housing panels is configured to be disposed within the first recess of the frame segment, and a third housing panel of the plurality of housing panels is configured to be disposed within the second recess of the frame segment.
 2. The housing of claim 1, wherein the frame structure comprises an additional frame segment, the additional frame segment comprises an additional protrusion formed on a third side of the additional frame segment, a third recess formed on a fourth side, opposite the third side, of the additional frame segment, and a fourth recess formed on the fourth side of the additional frame segment, wherein the first housing panel of the plurality of housing panels is configured to couple to the additional protrusion of the additional frame segment.
 3. The housing of claim 1, wherein the first housing panel is oriented crosswise to the second housing panel and the third housing panel in an assembled configuration of the housing.
 4. The housing of claim 1, wherein the frame segment defines an opening extending therethrough, the housing comprises a frame connector, and the frame connector comprises an extension configured to extend into the opening to couple the frame connector and the frame segment to one another in an assembled configuration of the housing.
 5. The housing of claim 1, wherein the frame segment partially defines a first housing section and a second housing section of the housing, the first housing section is an air handling unit, and the second housing section is a vestibule.
 6. The housing of claim 5, wherein the first housing panel is configured to extend between the first housing section and the second housing section in an assembled configuration of the housing.
 7. The housing of claim 1, wherein the second side of the frame segment is an exterior side, the first recess extends along a first lateral side of the frame segment, and the second recess extends along a second lateral side, opposite the first lateral side, of the frame segment.
 8. The housing of claim 1, wherein a surface of the protrusion is offset from an interior surface of the first side of the frame segment via a portion of the protrusion.
 9. The housing of claim 1, wherein the first housing panel is configured to directly couple to a first surface of the protrusion, the second housing panel is configured to directly couple to a second surface of the first recess, and the third housing panel is configured to directly couple to a third surface of the second recess.
 10. The housing of claim 9, wherein the first surface of the protrusion is oriented crosswise relative to the second surface of the first recess and the third surface of the second recess.
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